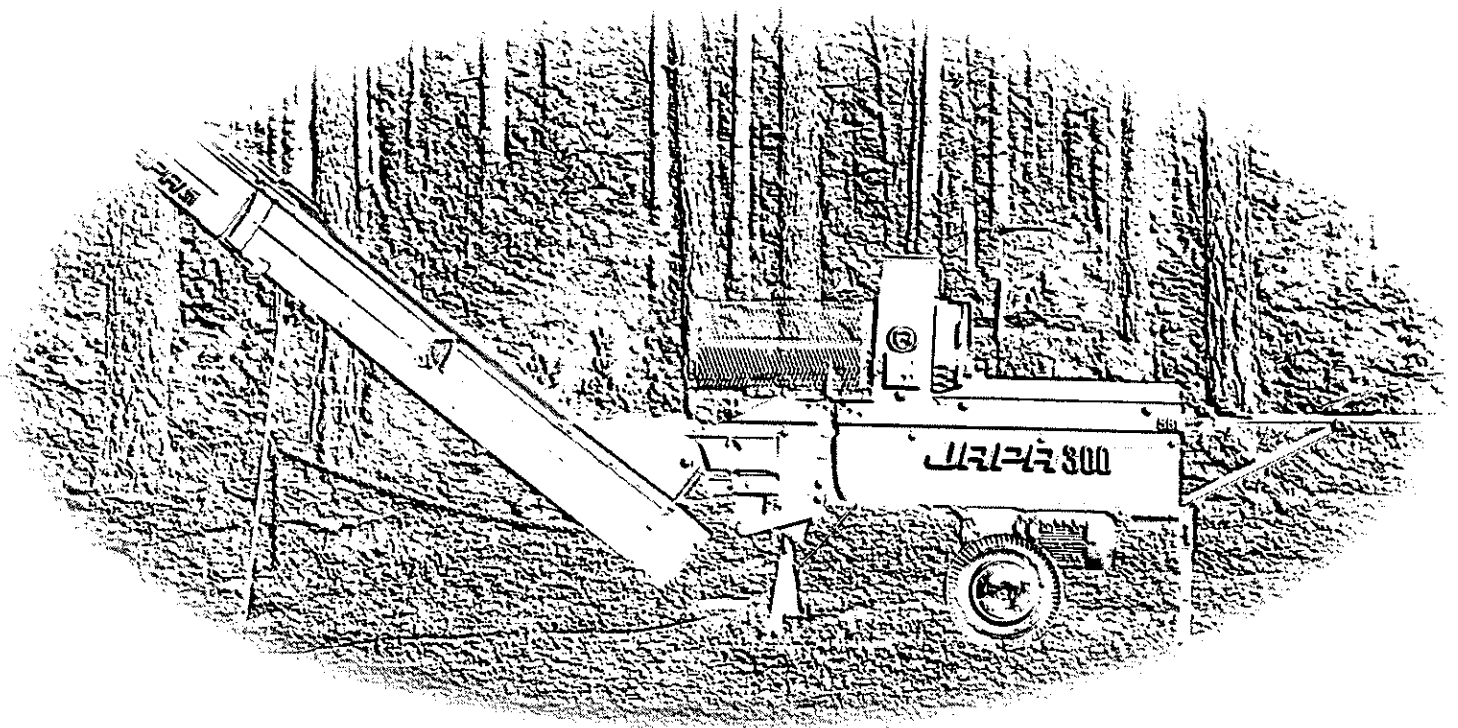


***JAPA* 300**

KÄYTTÄJÄN KÄSIKIRJA INSTRUCTION MANUAL



Valmistaja / Manufacturer: LAITILAN RAUTARAKENNE Oy
23800 LAITILA
FINLAND
www.japa.fi

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INSTRUCTION MANUAL

Laitila, Finland, 21.05.2007

FOREWORD

Instruction Manual

The instruction manual is intended as a guidance for an experienced firewood machine operator. Operating the machine requires average level of knowledge and skill.

Right to make changes

All instructions, illustrations and technical information are based on the latest level specifications at the time of the printing of this manual. The manufacturer will, however, carry on with the product development and will therefore retain the right to make changes, without any prior notice, to the design and construction of the machine which will improve the safety and quality of the product.

Machine Identification Plate and contact particulars

In order to ensure fast and efficient service in cases of spare parts orders or in breakdown situations, the Machine Identification Plate information should always be informed to the supplier of the machine or the service technician.

Mark down carefully the Identification Plate information on the appropriate space reserved on this page so that the info is always easily available when needed.

Type: **Japa Firewood Machine 300 TR5,6T, 300 E5,6T, 300 BE5,6T**

Serial No:

Manufacturing year:

If you are unable to diagnose the fault with the machine, contact the machine supplier who will diagnose the problem together with the manufacturer.

Supplier:

Company:

Address:

Tel:

Manufacturer's contact particulars:

Address: **Laitilan Rautarakenne Oy**

FIN-23800 Laitila Finland

www.japa.fi

Tel Int: **+358-2-8571 200**

Fax Int: **+358-2-8571 201**

We have manufactured a safe and reliable machine for you. You have the responsibility for its proper maintenance and for the correct use of the machine. This will ensure that the machine will serve you efficiently and reliably for a long period of time.

Familiarize yourself thoroughly with the machine Instruction Manual before the installation of the machine and before commencing the work with it.

Familiarize yourself also with the machine, its controls and emergency stops before commencing to work with it.

Keep the Instruction Manual always with the machine.

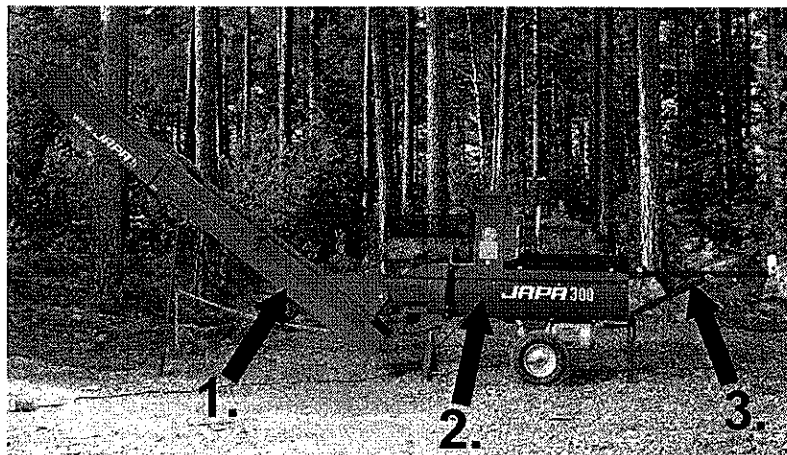
DESCRIPTION OF THE MACHINE

Japa Firewood Machine is a semiautomatic firewood cutting and splitting machine.

Use the machine only for this purpose! Japa firewood machine is a **one man** machine. Japa firewood machines are manufactured as tractor driven, electric motor and gas engine drive models.

Main parts

1. Conveyor
2. Firewood machine
3. Extension roller



Principle of operation

The usual power source for the machine is an electric motor. The machine drive input is coupled to the tractor PTO shaft by means of a U-J shaft, and the machine itself is supported by the tractor 3-point linkage. Cutting of timber takes place with a chainsaw and the splitting of sawn logs by a hydraulic splitter. Timber to be cut and split is placed on the feed table and it is then pushed against the log length limiter. The cutting handle is pushed gently until the log is cut through. After that **you have to hold the switch in the same position for a moment, until the log falls in its place in the splitting chute. This will ensure a successful split.** The cutting handle is then pushed down further, to its lowermost position, where it then triggers the log pusher equipped with hydraulic cylinder. The pusher then presses the log against the splitting knife.

Depending on the splitter knife version, the splitting of a piece of log can be done either to two, four or six parts. The height adjustment of the splitter knife is done by the control lever.

The chopped firewood is moved automatically by the conveyor directly to a transport load, to a pile of ready firewood further away, or directly into sacks. Moving of the chopped firewood further out increases the work safety as it prevents the chopped wood from piling up against the machine.

Accessories

The saw chain is driven by a hydraulic motor, which in turn is driven by a hydraulic unit. The hydraulic unit is powered by a tractor, electric motor or gas engine. The oil pressure needed by the machine hydraulic system for log splitting is provided by a hydraulic pump. Tractor driven Japa firewood machine is equipped with a gear system. U/J shaft is not provided as standard equipment. The electric motor drive version is equipped with a 4 kW electric motor and with an electric socket which can be connected to a 16 Amp, 5-pin extension cable. The machine has also an emergency stop system depending on the drive system applicable. Sack filling stand and firewood sacks are available as optional accessories.

Product line

JAPA 300 E5,6T - Electric motor driven model

JAPA 300 TR5,6T - Tractor driven model, tractor hydraulics

JAPA 300 BE5,6T – Gas engine

SAFETY PRECAUTIONS

!!! ATTENTION !!!

Read before the start-up of the machine

These safety precautions are of general nature. All other relevant safety and health regulations, traffic regulations during transport and common laws must be adhered to when handling the machine. Adhering to the rules and regulations will help to prevent accidents. Familiarize yourself carefully with the machine, with its coupling to the power source, its control and monitoring equipment and with the instructions before the installation and use of the firewood machine.

The machine may be operated only by a person who is familiar with it and with the relevant operating instructions of the machine. The operator must be in sound health, and it is prohibited to operate the machine under the influence of alcohol or any drugs.

Work site

- The operator work position, when the machine is running, is by the blade feed table
- Select a level and firm work site.
- Organize the site properly, keep it clean from unwanted materials.
- Ensure free passages in the working area during the operation.
- Ensure that the working area is not slippery.
- Ensure adequate general lighting of the work site.
- The Japa firewood machine is a **one man** operated machine. Make sure that there are no other people within the working site. If, however, there are more people working on the site at the same time, it must be ensured that they do not constitute a danger to themselves and to each other.

Personal protection and clothing

- Wear appropriate clothing. The clothing must not contain projections or loose parts which could get caught by the moving assemblies of the machine, thus causing accidents. Special clothing and personal protection used for common forestry work is suitable also for firewood machine operations.
- Use personal ear protection when the noise level exceeds 85 dB(A).
- Eye protection or safety goggles prevent eye damages caused by flying chips which may otherwise occur in spite of the appropriate guards.
- Non-slippery safety boots prevent slipping and protect the operator from foot injuries.
- Non-slippery gloves enable tight and firm grip even on a slippery piece of timber.

Before commencing the operation

- **FAMILIARIZE YOURSELF WITH THE MACHINE**
- **READ THE OPERATING INSTRUCTIONS**

Emergency stop

The electric motor driven firewood machine is equipped with an emergency stop which, when pressed, will stop all the machine functions. In tractor driven machines the emergency stop is arranged via a cord connection from the firewood machine stop lever on the top of the machine to the tractor engine stop lever, or to the PTO clutch control lever depending on tractor model in question. Make sure that the safety cord is always within the reach of the operator and test the emergency stop system before starting to work with the machine. The safety cord must be aligned in such a way that there is a clear sight all along the way of the cord. Always, when there has been an emergency stop situation, inspect the machine carefully. The U-J shaft drive must be uncoupled for the period of inspection. Before restarting of the machine, the safety cord must be reconnected. **Note that the stopping of the splitter by the stop lever does not stop the rotation of the chainsaw.**

Notes related to the operation of the machine

- Always check the amount of hydraulic oil from the dip rod before starting the engine.
- Lubrication of saw chain consumes hydraulic oil.
- Check that the saw chain of the saw is sufficiently lubricated.
- The amount of saw chain lubrication oil must be adjusted if the oil is cold or warm
- **In cold weather (below 0°C) it is advisable to “run the engine idle” for about 5-10 min before starting to saw.** In idle run the saw engine is running and the hydraulic oil circulates in the engine and becomes more flexible. If sawing is started immediately after start-up, the thermal relay of the engine may go off. The thermal relay recovers automatically in about 2 minutes and then the engine may be started again.
- The machine guards and protections are there for the safety of the operator. **Make sure the all the guards are in their respective positions** and in order. The machine is not to be operated if any of the safety guards have been removed, or if they are not in order.
- **Never touch the rotating parts.**
- **Test the functions of all the safety equipment and emergency stops.**
- Take extreme care when coupling the machine on or off the power source.
- Always, when commencing any service or repair work, **stop the machine**, turn off the motor, and in case of electric motor drive, pull out the plug from the socket.
- When leaving the machine, stop it.
- Always take special care when working with the firewood machine.
- Do not allow other people to approach the running machine (i.e. one man operated machine)
- **When the machine is running, be especially aware of the rotating blade and moving splitter.**
- The max rotating speed of the sawmotor is 6000 r.p.m. This speed must not be exceeded.
- With electric motor driven machine it must be ensured that the direction of the chainsaw rotation is correct. Maximum chainsaw speed also with this version is 21 m/s.
- The maximum pressure of the hydraulic system is 200 bar. The pressing force of the pusher is approx. 56 kN.
- If a piece of wood is stuck in the processor or conveyor, stop the machine at once.
- Before commencing the removal of the wood waste and sawdust, stop the machine, turn off the motor, and in case of electric motor drive, pull out the plug from the socket.
- **Clean the machine only after it has stopped completely.**
- Keep the machine emergency stop system on stand-by and clean.
- **Do not cut timber in a bunch.** Cut only one trunk at a time.
- **Do not saw timber with nails in it.**
- Take special care when cutting twisted or long branched trees
- Ensure that the timber is pressed tight against the cutting table and that it is well supported before commencing sawing.
- **Make sure that the timber cannot wedge itself against the blade.**
- Do not attempt to cut too long trunks. A too long piece of timber will press against the blade.
- Be aware of your back, move the timber from side, do not pull the trunk by the blade end. Remember the right grip.
- The max diameter of a trunk to be cut is 300 mm and the minimum 20 mm.
- Remove the chopped wood from the vicinity of the machine before it starts piling up against it.
- **Service the machine in due course.**
- Cover the chainsaw when not in use. Be aware of sharp chain saw.
- When opening the sawing guard, stop the machine, turn off the motor, and in case of electric motor drive, pull out the plug from the socket.
- The splitting zone is safeguarded with an opening guard. When the guard is opened, the splitting action taking place is carried out to completion. Do not open the guard until the pusher has completely stopped.
- The pusher may move when starting the machine and therefore the guard must be closed prior to start.
- When the splitting zone guard is opened, the operating lever is mechanically locked up to the upper position keeping the saw chain from rotating and preventing the usage of the cutting and splitting functions.
- Cutting and splitting functions do not work until the splitting zone guard is closed.
- **Attention!** Wheels of this device are suitable for short distances only, e.g. for its relocation within the yard. Maximum speed is approximately 40 km/hour.
- **Persons under the age of 18 are not allowed to operate the firewood machine!**

WARNING LABELS

There are a number of compulsory and important instruction and warning labels on the machine. By following these strictly, accidents and breakdowns will be avoided. Locations and notifications on these warnings and instructions are presented in the following illustrations.



Identification Plate and CE-mark

JAPA

Japa stickers

MAX R. P. M - 400

Maximum allowed PTO speed



Arrow on the blade guard indicating the direction of rotation.

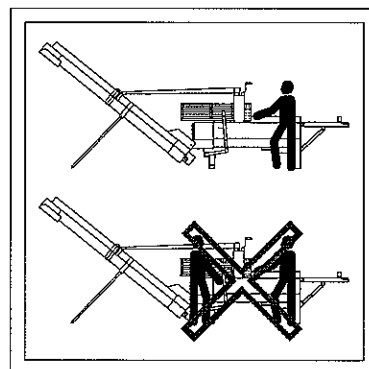
U-J shaft direction of rotation



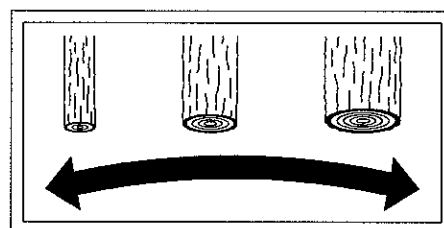
On top of the blade guard:
BE AWARE OF ROTATING BLADE



Wear personal ear and eye protection.



Only one user sticker



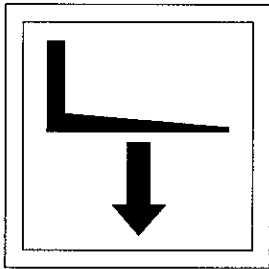
When splitting big trees, move the lever to the direction of the arrow. Then the axe rises and splits the trees to four or six parts.



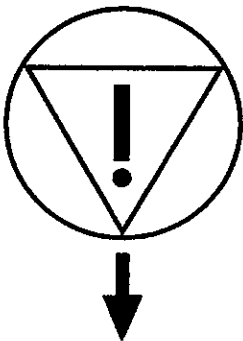
Log length metering sticker



Warning stickers of the conveyor:
Do not touch the end rolls when the conveyor belt is rotating.



Positions on the firewood harvester and conveyor for lifting with a forklift truck.



Hätä – seis

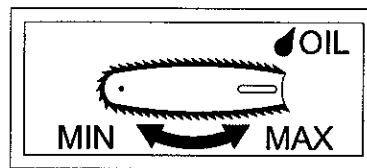
Stopping of tractor engine: pull in the direction of the arrow
Electric drive: press the button



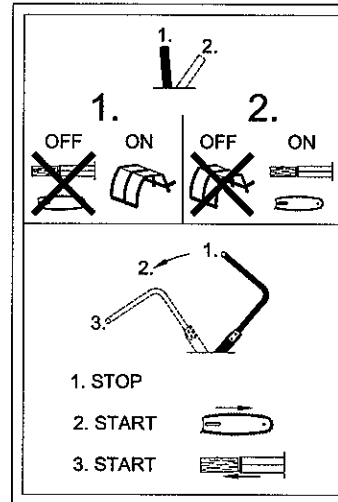
The oil in the hydraulic system



Read the Operator Manual sticker



Saw lubrication regulating valve



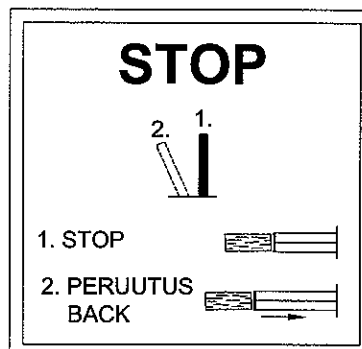
Safety locking sticker:

- In position 1 splitting and cutting are switched off. You can open the protection of the splitting area.
- In position 2 splitting and cutting are switched on. You cannot open the protection of the splitting area.

The locking handle can be put in position 1. only if the handle of use is in upper position and the pusher has stopped. The handle can be put into position 2. only if the protection of the splitting area is closed.

Operating lever

- Position 1. neutral .
- Position 2. the saw chain starts
- Position 3. the splitting function starts



Stop / Back sticker

- In position 1. the pusher stops when the stop lever is turned to middle position.
- In position 2. the pusher reverses when the stop lever is turned to the left.

INSTALLATION INSTRUCTIONS

The firewood machine should be placed by the log pile to be sawn and chopped so that passages and the working site by the machine are free. Check before starting the machine that all the fastening screws are tight and that all the guards are in their proper places. Before commencing the work, check also that the machine is firmly located. Clean the chainsaw, check its condition and ensure it is sound before starting the machine. Check also the condition of the control and safety equipment before the start-up, and if there are faults, rectify them before you start the machine.

U-J shaft drive

In U-J shaft driven installation the firewood machine is first fastened to the tractor 3-point linkage. It is then centered and locked with the linkage stabilizer bars. The length of the U-J shaft must be checked. Note also that machine end of U-J shaft protective guard must extend at least 50 mm over the shaft. A non-contact check must be carried out when raising the 3-point linkage to its upmost position. Shorten the U-J shaft if necessary. **The speed of the U-J shaft must not exceed 400 r.p.m.** For safety reasons this speed must not be exceeded. When the U-J shaft is not coupled to the tractor, it can be lifted onto holding bracket by the machine side.

Tractor:

- The U-J shaft can only be coupled when the tractor PTO clutch is disengaged. Never allow the uncoupling of the U-J shaft relying only on the disengagement of the PTO clutch.
- When starting the tractor engine, the PTO shaft must be disengaged.
- Before engaging the PTO shaft clutch, make sure that there is nobody within the danger zone.
- Always use the correct PTO speed.
- The PTO shaft must be disengaged when not in use.

U-J shaft:

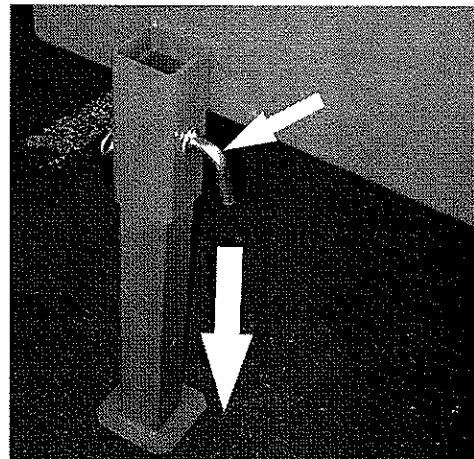
- Before engaging the the PTO shaft and during the PTO use, make sure that there is nobody within the danger zone.
- Never use anything else except a faultfree, protected and approved U-J shaft.
- If a U-J shaft is shortened, take care of its cleaning and lubrication.
- If a U-J shaft needs to be shortened, follow the manufacturer's instructions.
- In an installed U-J shaft, the locking pin must be inserted and locked in the locking groove on the driven shaft. Check that the shaft is securely in its proper position and locked.
- Connect the shaft guard security chain so that the guard cannot rotate.
- For the uncoupled U-J shaft there is support bracket by the machine side. Never hang the shaft from its security chain.

Electric motor drive

- As an electric motor driven version the machine is equipped with a 400 V, 4 kW electric motor, which can be connected up a power source by means of a 16 Amp, 5-pin extension cable. The machine is equipped with an emergency stop system. Check that the direction of the rotation is correct.
- If the motor runs the wrong way around, stop it immediately, remove the plug from socket, change the two phase wires around and try again (**leave the electric installations to an authorized electrician**).

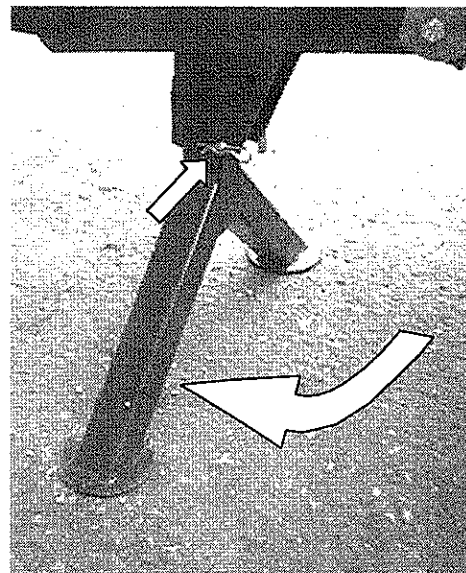
Installation of the back support

- Remove the bin and dowel of the support and bring the support down.
- Put the dowel back to the lower hole and put the bin back.
- When you want to change the engine back to the transportation position, do the same procedure in opposite order.



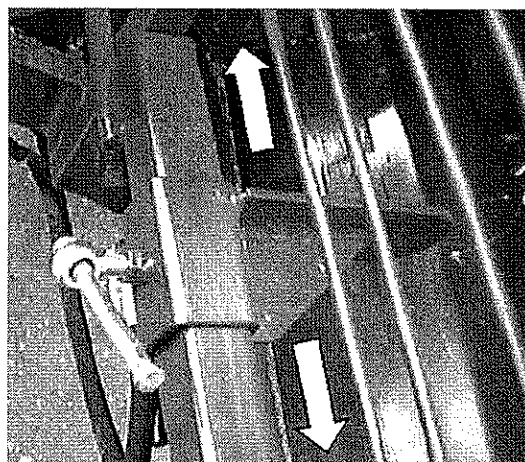
Installation of the support

- Remove the bin and dowel of the support and bring the support down.
- Put the dowel back to the lower hole and put the bin back.
- When you want to change the engine back to the transportation position, do the same procedure in opposite order.



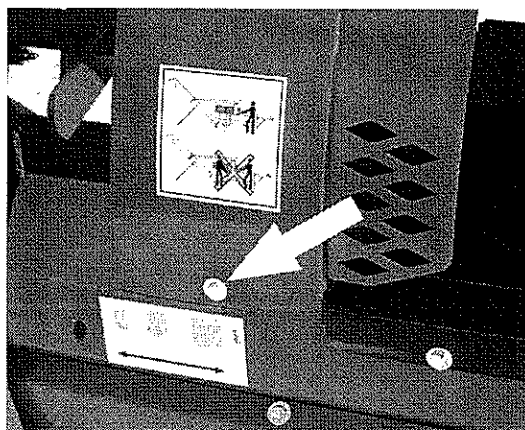
Adjusting log length

- Loosen the clamp of the limiter and adjust the limiter to a desired distance.
- Tighten the limiter.
- If the log length limiter has been adjusted for a long log, but the log to be split is however short and the operating lever is returned quickly back to its neutral position, the limiter goes in front of the log. In this case the operating lever must be held to its lowermost position until the log has been splitted.



Installation of the saw chain guard

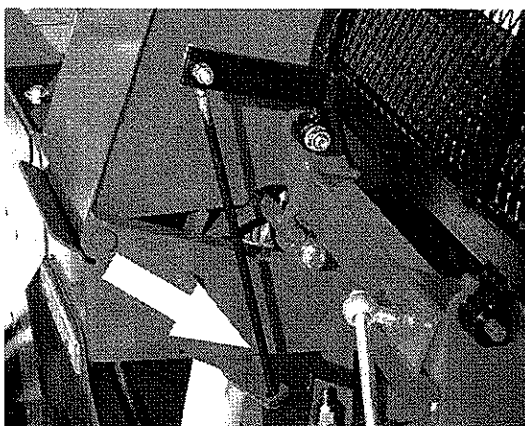
- Before starting the machine check, that the saw guard has been installed.
- Install the fastening screw in front of the saw guard



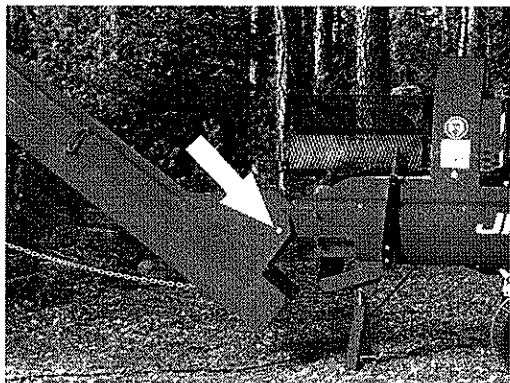
Installation of the splitting zone guard

Before taking the machine into use, install the splitting zone guard.

- Check the saw axle locking when the splitting zone guard is open



Installation of the conveyor

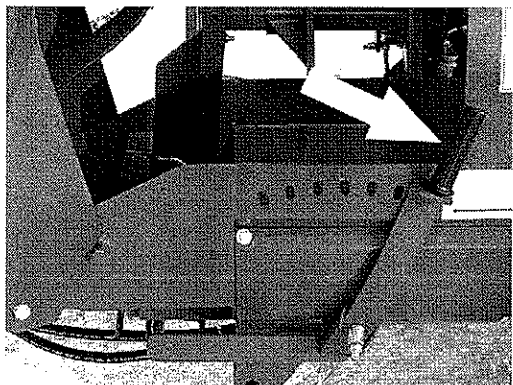


Before the firewood machine is taken into use, a discharge conveyor must be fitted to the machine. Install the fastening bolts of the conveyor

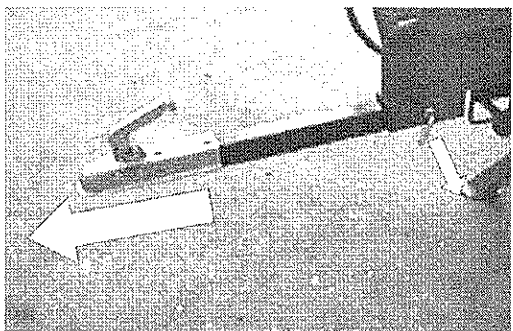
Putting the conveyor to working position



1. Conveyor in its transportation position.



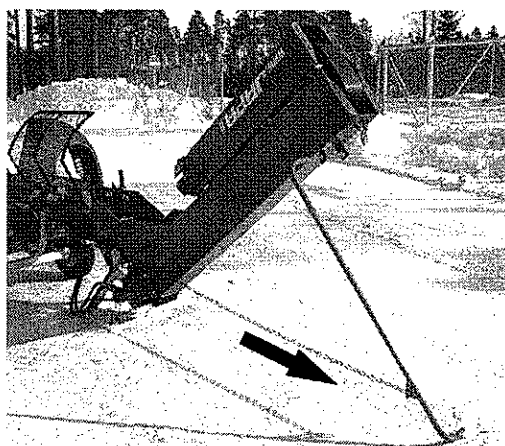
2. Make sure that the splitting blade is in the upper position.



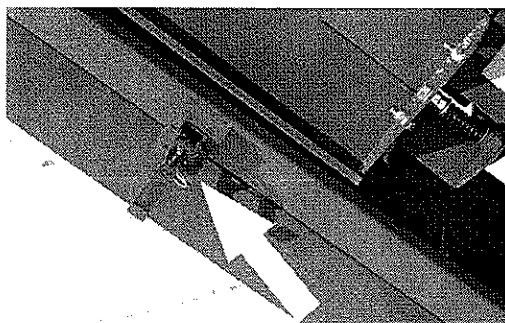
3. Remove the drive shaft of the firewood machine



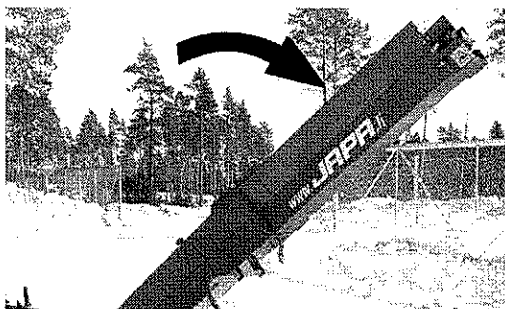
4. Remove the support leg and secure it to the other side of the conveyor. Put the locking chain of the leg in a more loose position.



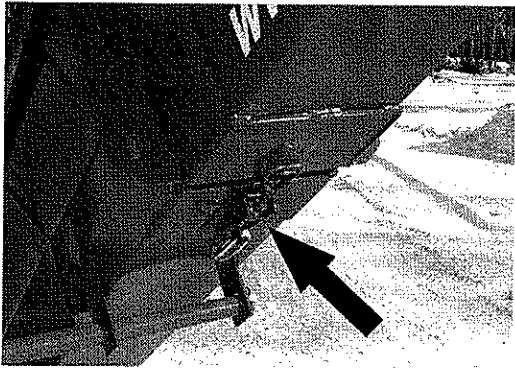
5. Lower the conveyor, until supported by the leg.



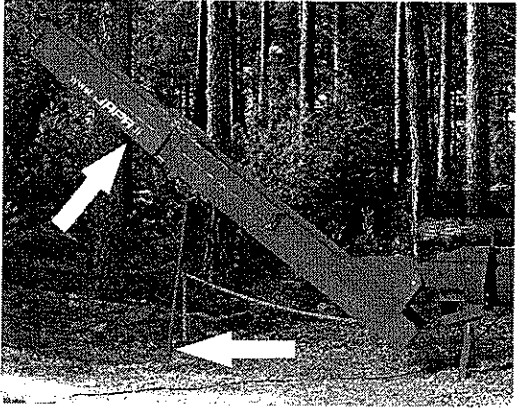
6. Open the locking hook at the side of the conveyor



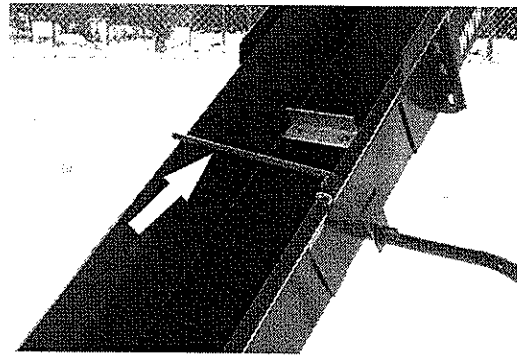
7. Straighten the upper part of the conveyor.



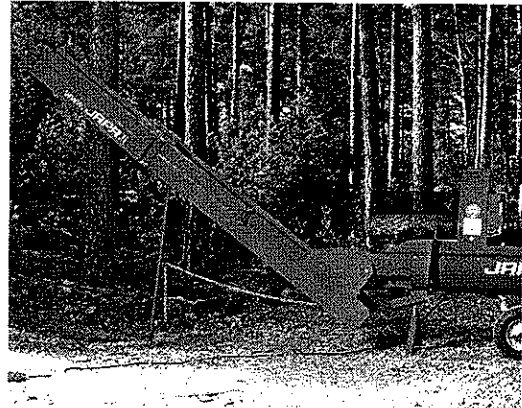
8. Lock the upper part of the conveyor to the lower one by means of the locking hook at the base. Secure the locking with a circle cotter.



9. Lift the conveyor up in a straight position to the appropriate height and tighten the leg chain.



10. Turn the belt support aside.



11. Conveyor in its operating position. Lower the conveyor down and make sure that the support is firmly in place and the chain is tight. If the leg is too upright or the conveyor should be lower, the chain length can be adjusted.

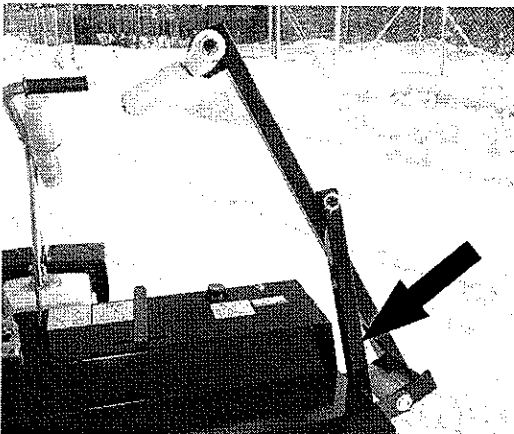
Putting the conveyor to transportation position

Do the same procedure in opposite order.

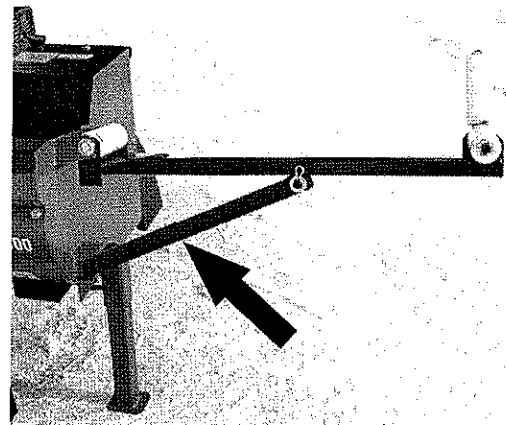
Always make sure that the protection of the splitting area is open and the splitting blade in the upper position when placing the conveyor in the transportation position! Keep your foot on the support leg as you lift the upper part of the conveyor to transportation position.

Extension roller handling

Extension roller to operating position



3. Remove the support leg and secure it to the other side of the roller.



4. Lower the extension roller down and install the support to its place

Extension roller to transportation position

Do the same procedure in opposite order.

SERVICE INSTRUCTIONS

Service instructions of the saw chain and the chain blade

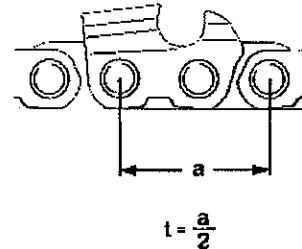
Sawing takes place effortlessly when the saw chain is sharpened correctly. The well sharpened saw chain penetrates into the wood with a little of effort. It is very uneconomical to work with a dull or damaged saw chain. Clean and check the saw chain and make sure that there are no splits in the chain joints and that all the chain rivets are sound. Damaged or worn out chain must be replaced with a new one.

Use only special saw chain sharpening files!

Remember the correct tension of the saw chain!

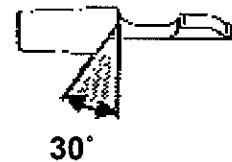
The saw chain has the following chain division: $t = 0.325''$

For checking of the division, t = distance between 3 rivets divided by 2



Constant sharpening angle (degrees):

- The angles must be the same for all the teeth of the saw chain. If angles vary, the saw chain will move unevenly, it will wear down more quickly and it may even break.
- As these requirements can be fulfilled only after sufficient experience, **please do use filing guide!**



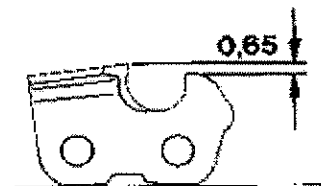
All the cutting teeth must be of equal length.

- If the teeth are of uneven lengths, also the tooth heights are uneven and this will cause uneven rotation of the saw chain, which eventually causes splits in the chain.
- All the teeth must be filed according to the shortest tooth. This could be time consuming, and the best way to do the job is to get it done in a workshop by means of an electric sharpening device.



Filing of the depth regulator

- Depth regulator determines the penetration depth into the wood and thus the thickness of the chip. Depth regulation distance becomes smaller along with filing.
- Check the depth regulation distance by means of a reduction diagram. If required, file with a flat or triangular file.
- The distance between the depth regulator and cutting edge is = 0.65 mm.
- When cutting coniferous trees, it could be 0.2 mm bigger, except when the outside temperature is below zero degrees centigrade.



After sharpening:

- Clean the saw chain thoroughly, remove any stuck chips and filing dust and submerge the chain into oil bath.

When the saw chain is not used while:

- Clean the chain with a brush and store it in oil/paraffin oil bath.

Servicing the blade

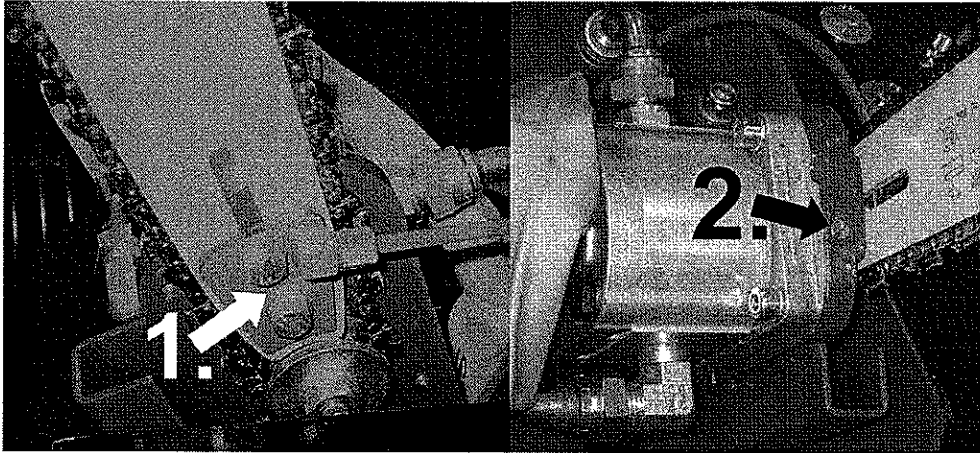
Reverse the blade and file from its side, and clean the groove when necessary.

Replacement and tightening of the saw chain

Turn off the motor before performing maintenance work to the saw chain!

- Loosen the hex screw at the front of the saw bar cover and turn the cover open.
- To tighten the saw chain loosen the retaining nuts (1.) of the saw bar and turn the tightening bolt (2.) clockwise.
- If you need to remove the saw chain, turn the tightening bolt (2.) counter-clockwise until the chain becomes loose.
- The chain should be tightened, so that it would not hang loose under the saw bar.
- Finally retighten the retaining nuts (1.) of the saw bar.

Check the chain tightness regularly.



Bearing lubrications

Bearings used are sealed for life and they do not need to be lubricated necessarily. By the end of the running season it is, however, beneficial to add a little lubricant to the bearings. That can extend their service life. If too much lubricant is forced into the bearing, the seal can be damaged.

Check and change of hydraulic oil and saw chain lubrication

- The oil grade is e.g. Mobil Flowrex 46
- The amount of oil is checked from the dip rod in the cap
- The minimum oil quantity is 20 l (the dip rod must get wet)
- Emptying of the reservoir takes place by disconnecting the suction hose
- Oil change should take place once a year, as a principle

Conveyor belt adjustment

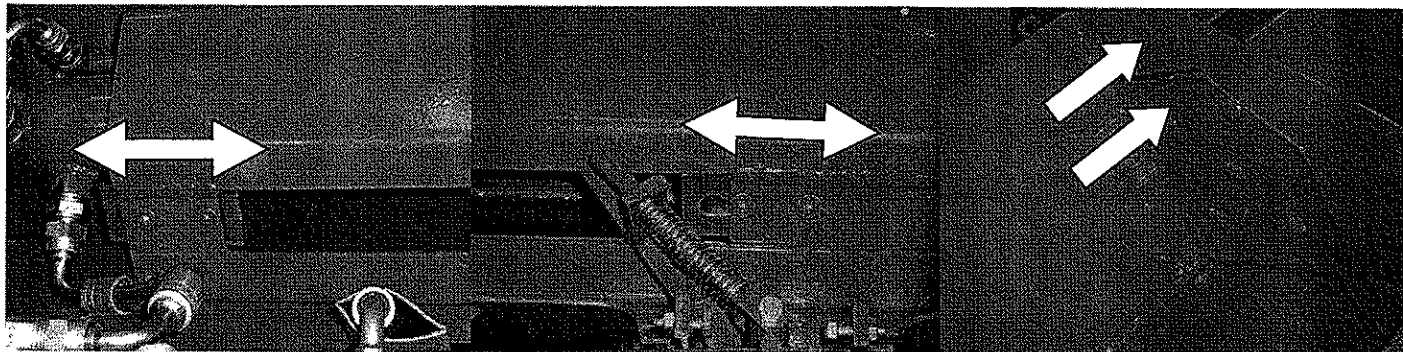
- Flat belt is tensioned by the upper end hex screws.
- The same screws are used to adjust the belt to run centrally.

Adjustment of the splitter valve

Adjustment of stroke

It is most important to remember that the cylinder must never bottom, i.e. it must stop approx. 10 mm before the end.

The stroke is adjusted by moving the two assembling stoppers of the splitting device frame. Loosen two hex socket head screws and move the stopper gradually. Tighten the screws. Find first the bottoming part and then move the stoppers in order to stop the cylinder about 10 mm before the end. Test run the pusher a few times.



1. The left side stopper adjusts how far the pusher can go.
2. The right side stopper adjusts how much the pusher is pulled back.
3. Stroke is correct when the pusher end plate is on the same level with the pullback plate.

Valve adjustment instructions

Adjustment screws (1) are the joint swing limiters. Adjustment screw settings take place as follows.

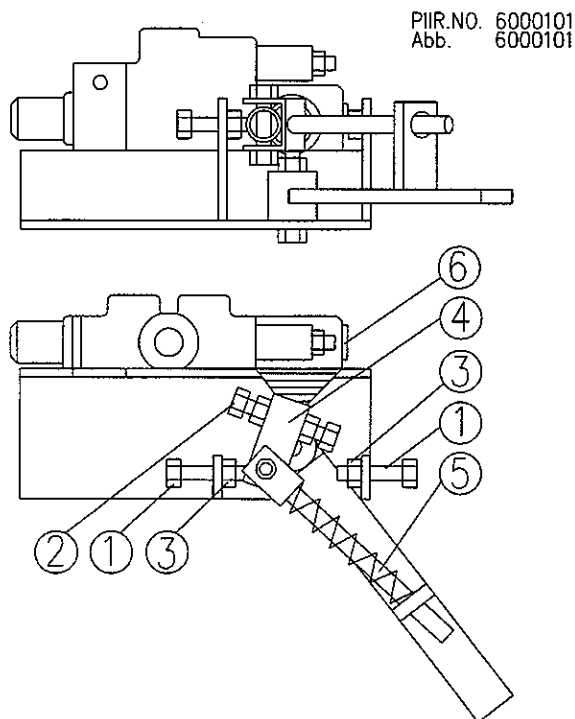
1. The joint sleeve linkage (4) is released to its end position.
2. The adjustment screw (1) is turned out until the tip of the screw does not reach the joint sleeve any more.
3. The screw is turned in until the screw tip pushes the joint sleeve 2 mm forward.
4. The bolt locking nut (3) is tightened.
5. Same actions are repeated on the other side.
6. Screws for the joint sleeve must be tightened.

Now the limiters are set correctly and they will stop the joints at turns, and thus the internal linkage of the valve will be spared from the shocks caused by turns.

The valve neutral position is adjusted by the adjustment screw (2).

When turning the adjustment screw out, the pusher moves backwards.

When the screw is turned in, the pusher moves forwards. Adjust the screw out or in until the pusher, when stopped in the middle of its operating phase, does not move in either direction. Grease should be brushed on the shaft pin (5), when necessary.



HYDRAULICS

The splitter is operated hydraulically. The flow capacity of the hydraulic pump is 30 l/min. The system pressure is 200 bars. The pusher pressing force is 40 kN. Check, with engine running, the functioning of the hydraulic and stopping systems before commencing the work.

- When the firewood machine is running, there is a high pressure in the hydraulic system. Leaking high pressure hydraulic oil can penetrate the skin and cause serious injuries. There is a risk of an injury even when tracing leaks in the system. In case of an injury, contact a doctor immediately (danger of inflammation).
- Be careful with all hydraulic components. Use proper protections and accessories.
- Check the hydraulic hoses on regular intervals and renew them if they are worn out or damaged. All the replacement hoses must conform to the appropriate DIN or other standards.

Note when replacing hoses:

- Always before disconnecting a hydraulic coupling, make sure that the hose is free of pressure.
- When replacing hoses, check that they are not twisted or that they do not get twisted when the machine is operating.
- Do not over tighten the hose coupling nut.
- Check the position of the hydraulic coupling.
- Always use the correct size hose clamps.
- Tighten the outside threaded (males) couplings before tightening the coupling nuts.
- Do not use locking or sealing compounds.
- Always use correct size spanners.

Discard the old hose and hose coupling if:

- Hose has loosened from its coupling (discard also the coupling)
- There is split in the hose or the hose leaks (also when the hose gets wet due to an inside seepage).
- Hose surface has worn down to reinforcement layers.
- Hose contains a flat area or has a permanent change of form.
- There are signs of fire or the hose surface is cracked.
- Hose manufacturer does not warrant the hoses for more than five years.
- Coupling is corroded, or contains a permanent change of form.
- Coupling leaks.

Press fit couplings can only be used once.

Protection against oil and grease

Oils and greases can contain additives which in some cases can be harmful when there has been a repeated skin contact. Follow the appropriate safety instructions when handling these substances.

- Always wear protective clothing, use appropriate hand cream or oil resistant gloves when handling oils and greases.
- Spreading of oil mist to the working site or oil spill to walking passages will cause a slipping danger.
- Avoid skin contact with oils and greases.
- Never use oil or grease for cleaning the hands.
- Change oily clothes as soon as possible.
- No oily rags etc. in the pockets.

Waste oil must be collected and when discarding of it, the appropriate rules and regulations must be followed.

STORING INSTUCTIONS

- Clean the machine thoroughly inside and outside. Dirt accumulates moisture which causes rust.
- Washing can be done with pressure washer, but please, avoid directing the water jet straight on the bearing and other parts which can easily be damaged.
- Lubricate all joint with grease.
- Unpainted, unprotected and worn parts should be treated lightly with oil.
- Do not store machine next to artificial fertilizers.
- Store the machine on firm, even ground. The discharge and feed conveyors can be left in transport position.
- Lubricate the chain flange.

DISCARDING THE MACHINE

- The owner, persons or company, owning the machine is totally responsible for discarding the machine.
- Hoses, oils, rubber and plastic parts are to be disposed probably according to the directions of the authorities.
- Iron and other metal parts are to be recycled through machine stripper bays to be reused.
- For discarding the product and handling the waste involved, there are national laws and orders in every country which are to be followed.
- Detailed information concerning the waste disposal and stripping can be obtained from local authorities for the environment.

TECHNICAL SPECIFICATION

Firewood machine Japa 300

Weight of firewood machine.....	280-310 kg
Weight of conveyor.....	75 kg
Lenght in transport position	2780 mm
Widht in transport position	1030 mm
Height in transport position.....	2450 mm
Saw flange	13"
Chainsaw	0.325" - 56 links
Maximum lenght of log.....	500 mm
Maximum log diameter	300 mm
Electric motor	4 kW
Pusher pressing force.....	56 kN
Pusher working movement	19 pcs/min
Hydraulic pumps capacity.....	30 l/min and 4.8 l/min
Hydraulic oil pressure.....	200 bar
Oil tank capacity.....	27 l
Warranty period	1 year

Noice level of the machine

Noise pressure equivalent level 89,5 dB (A) +/-3 dB

Noise effect level 100,5 dB (A) +/- 1,5 dB

Vibration

The weight efficiency value of the accelatioeton force by the operator arms 1,19 m/s.

SERVICE SCHEDULE

Service item	Action	As needed	Daily	6 months / 100 hrs	12months / 500 hrs
Gear box oil	Check Change		X	1. X	Next X
Hydraulic hoses	Check Change	X	X		
Hydraulic oil	Check Change		X		X
Oil filter	Change			1. X	Next X
Release mechanism	Check Lubric.	X	X		
Emergency stop	Check at start				
Chainsaw and saw flange	Clean & Sharpen	X			
Electric equipment	Check		X		
Motor	Check & Clean		X		

Change of hydraulic oil:

Open suction pipe for
the the used oil.

Gear box oil change:

Suction removal of oil or removal of
removing drive.

TROUBLESHOOTING

TROUBLE	POSSIBLE REASON	CORRECTIVE ACTION
Chain saw makes contact with guard.	Saw shaft not aligned properly.	Open the bearing support screws and realign the shaft. Tighten the screws.
Splitting does not function	No oil, or too little oil	Stop immediately, add oil
Splitting does not function	Handle does not move down enough	Adjust triggering sensitivity
Splitting does not function	Valve adjustment has changed	Adjust the triggering to middle position from the valve
Splitting does not function	Litter in triggering system	Clean the triggerin system
Splitting does not function	Oil too cold	Let oil circulate on free circulation a few minutes
Splitting does not function	Hose broken or leaks	Replace the hose
Splitting does not function	Triggering system frozen	Clean the machine always after you have finished working
Slow or powerless splitting movement	Oil too cold. No oil or too little oil	Let oil circulate on free circulation for a few minutes. Add oil. Adjust valve to middle position
Log does not split	Incorrect knife position	Adjust knife height
Log does not split	A big knot at splitting point	Stop the machine, open the safety guard, turn the log, close the guard
Log does not split	Exceeds the splitter max limit	Max thickness 30 cm
Log does not split	Oli pressure has fallen down	Check the hydraulic system
Oil heats up too much	Too little oil / cylinder bottoms and oli pressure remains on, oil circulates via pressure relief valve.	Add oil,/adjuste triggering System and cylinder stroke and swing. Check the oil pump.
Log lifts up when splitting	Twisted or knotty log	Check the pusher stroke
Electric motor does not start	Emergency stop button down	Pull emergency button up
Electric motor does not start	Blown fuse	Replace fuse
Electric motor rotates the wrong way round	Two phase wires the wrong way round	Change the two wires in the plug the right way round (a job for an electrician)
Conveyor belt runs on one side	Adjusment has changed	Adjust the return roller at the end of the conveyor. Test driveafter adjustment
Log gets stuck at the splitting knife	Wrong stroke of the splitter function. Dull blade	Extend the srtoke. Sharpen the knife
Emergency stop does not function	Safety cord not tight or disconnected	Tighten or reconnect safety cord
Log collides with conveyor belt	Conveyor is too upright	Lower the conveyor slope angle (longer conveyor)

WARRANTY CERTIFICATE

Type No:

Serial No:

This Japa product is covered by a warranty valid for one year from the date of purchase of the product. Warranty covers manufacturing and material faults of the product. Parts and components delivered to replace the faulty parts will not be charged to the customer.

The warranty does not cover:

- a machine which does not have the original Identification Plate
- damage caused by abnormal use
- damages caused by natural wear and tear
- consumer parts such as V-belts, sawchains etc.
- wash, cleansing, oil and petrol
- normal adjustments, operating instructions, service and maintenance

Warranty is not valid during the warranty period if:

- the machine has not been used or serviced according to the instructions included in the delivery
- the ownership of the machine has been transferred to a third party

The shipment costs of a machine or a component to be repaired and the labour and travel costs of the mechanic are the responsibility of the owner of the machine. Possible damages which occur during the transport are not covered by the warranty.

This warranty expresses and limits our whole responsibility and it replaces and/or substitutes all other implied or expressed warranties and responsibilities which otherwise may apply.

In a case of a request for reimbursement, based on valid warranty, the Buyer (i.e. the owner) must advise the machine model code, serial number and appropriate details of the purchase receipt, indicating a.o. the date of purchase of the Japa machine in question. The reimbursement request should be presented to our authorized retailer.

All subjects raised under this warranty must be agreed beforehand with the Seller (i.e. the Supplier) or the Manufacturer of the machine before any action is taken.

LAITILAN RAUTARAKENNE OY
23800 LAITILA

www.japa.fi
TEL. +358-2-8571 200
FAX. +358-2-8571 201

***JAPA* PRODUCT**

CERTIFICATE OF CONFORMITY

(According to EU conformity certification directive 98 / 37 / EEC - Attachment A II)

(Name of the manufacturer)
(Address of the manufacturer)

LAITILAN RAUTARAKENNE OY
Kusnintie 44
FIN - 23800 Laitila, Finland
www.japa.fi

(Telephone Nr of the Manufacturer, Int:) +358/ 2/ 8571 200
Fax Int: +358/ 2/ 8571 201

It is thus assured that the product to be sold:

JAPA Firewood Machine, model 300 TR5,6T, 300 E5,6T, 300 BE5,6T

Optional equipment:

Machine serial Number:

is in accordance with the machine safety directive 98 / 37 / EEC and the changes made to it, enforced by the decree No 1104 / 99 of the Finnish Government. The machine is in accordance with the EU directives applicable and in accordance also with the corresponding national regulations:

.....

The harmonized standards and proposals for standards as applicable has been adhered to in the design and construction of the machine:

.....
.....

The product described in this certificate has been tested in the usual operating circumstances.

Place and date: In Laitila.....21.05.2007...



Signature:

Matti Piela. Managing Director
(clarification of the name and position)

REGISTRATION FOR WARRANTY

Date / 20.....

Product:

Machine serial Number

Optional equipment:

.....

.....

.....

.....
Buyer / Operator of the machine

.....
Address

.....
Supplier

.....
Company

Note concerning the Instruction Manual:

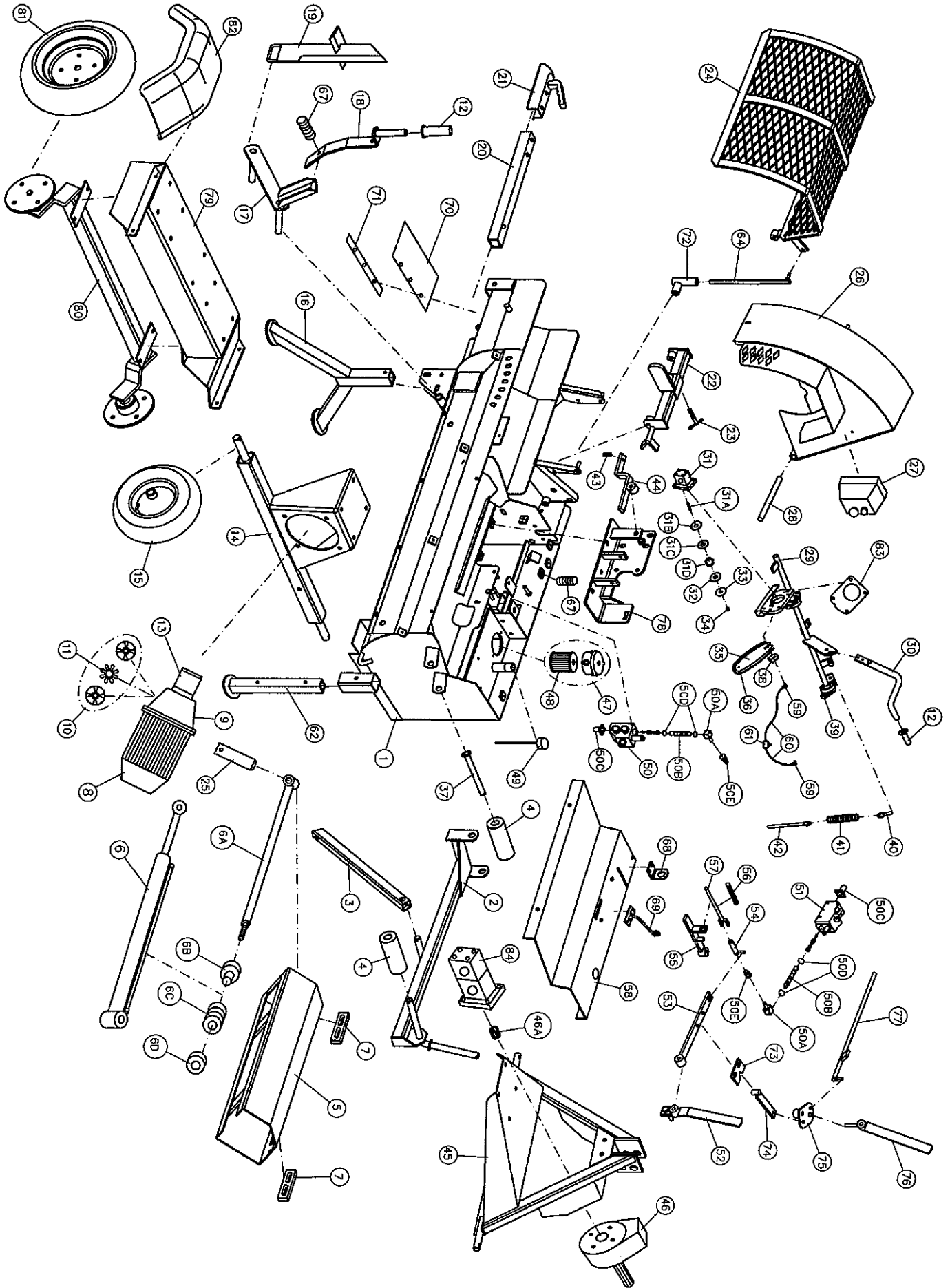
The appropriate machine Instruction Manual has been delivered with the machine, and the Buyer acknowledges that he has familiarized himself with the Manual before commencing the operation with the machine.

.....
Signature

The buyer will fill out this part and send it to the manufacturer/sales representative.

SPARE PARTS

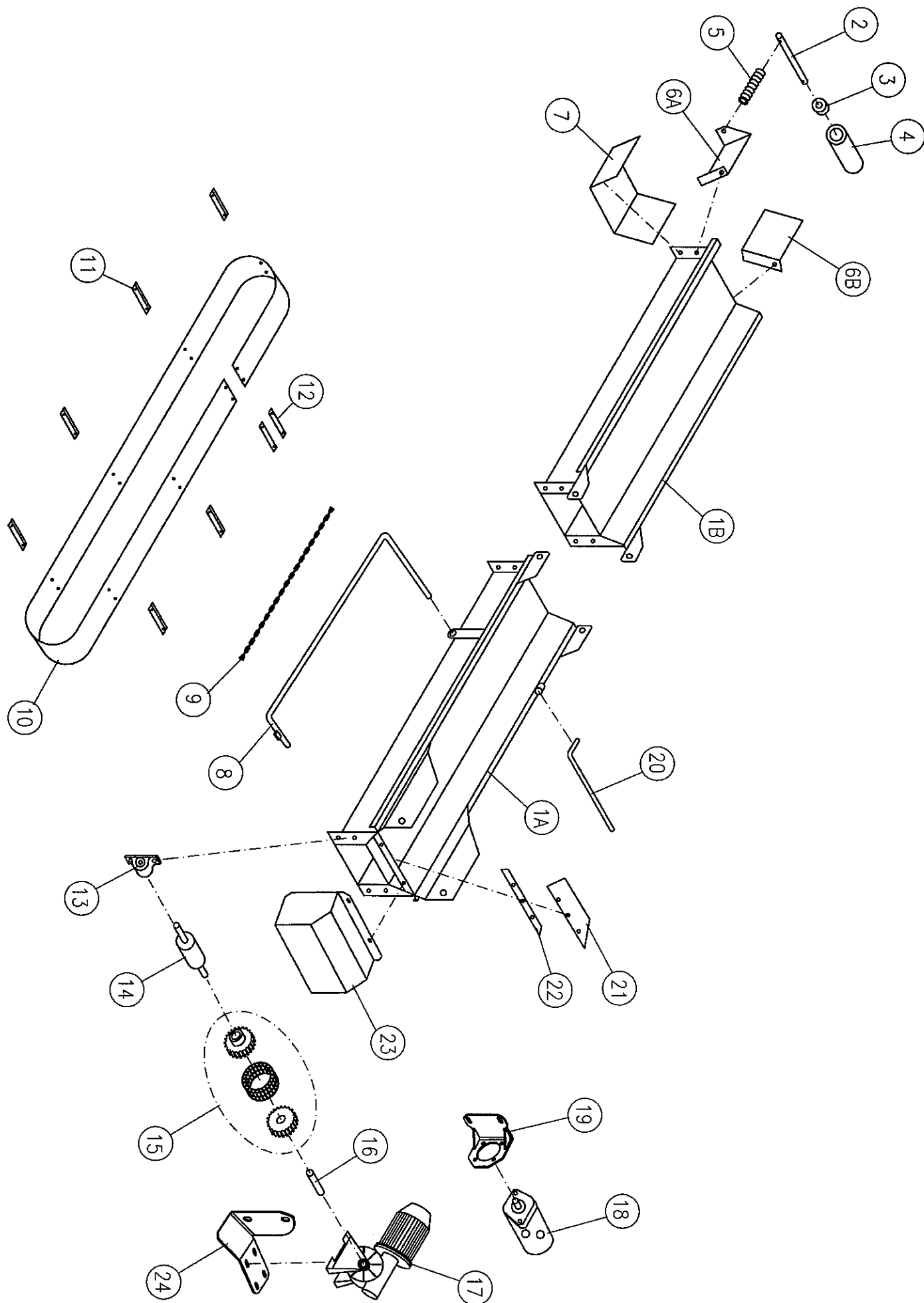
Japa 300 firewood machine explosion view



Japa 300 firewood machine spare parts list

Part	Part name	Pcs	Part	Part name	Pcs
1	Frame.....	1	43	Spring	1
2	Extension roller.....	1	44	Release stop	1
3	Support.....	1	45	Lifting triangle (only tractor-driven)	1
4	Roller.....	2	46	Gear (only tractor-driven)	1
5	Pusher	1	46A	Gear sleeve (only tractor-driven)	1
6	Cylinder.....	1	47	Hydraulic oil filter.....	1
6A	Piston rod.....	1	48	Filter element	1
6B	Box.....	1	49	Oil tank filling neck and dipstick	1
6C	Piston	1	50	Valve.....	1
6D	Rod seal	1	50A	U-joint casing	2
7	Turner stop.....	2	50A.1	U-joint pin.....	2
8	Electric motor, 4 kW	1	50B	Spool.....	2
9	Electric motor flange.....	1	50C	Protective case.....	2
10	Clutch	1	50D	Spool shaft O-ring	4
11	Clutch rubber.....	1	50E	Chain flange clamp	2
12	Handle.....	2	51	Valve.....	1
13	Hydraulic pump	1	52	Stop lever.....	1
14	Axle	1	53	Stop lever bar.....	1
15	Wheel.....	2	54	U-joint sleeve	1
16	Support.....	1	55	Tappet.....	1
17	Splitter knife lifting rod	1	56	Spring	1
18	Splitter knife handle.....	1	57	U-joint fork.....	1
19	Splitter knife	1	58	Cover	1
20	Drive shaft.....	1	59	Sawchains lubrication angle	
21	Drive clutch	1		male union	2
22	Log length limiter.....	1	60	Sawchains lubrication hose.....	1
23	Locking screw.....	1	61	Sawchains lubrication controller	1
24	Splitter guard.....	1	62	Support	1
25	Tap of cylinder.....	1	64	Drawbar	1
26	Blade guard.....	1	67	Spring	2
27	Electric starter	1	68	Leg lug	1
28	Blade guard shaft	1	69	Emergency stop (only tractor-driven). 1	
29	Saw shaft	1	70	Rubber of the debris rack	1
30	Saw shaft operating lever	1	71	Fixing iron for the rubber	1
31	Chainsaw motor	1	72	Connecting sleeve.....	1
31A	Chainsaw key.....	1	73	Adjustment piece.....	1
31B	Chainsaw seal.....	1	74	Locking of reversing	1
31C	Support plate to seal.....	1	75	Wobbler	1
31D	Lock ring.....	1	76	Safety locking handle	1
32	Drive pulley.....	1	77	Intermediate rod of splitting protection1	
33	Guide plate for drive pulley	1	78	Valve frame.....	1
34	Lock nut for drive pulley.....	1	79	Axle mounting (only bigger axle)	1
35	Saw flange	1	80	Axle (only bigger axle).....	1
36	Chainsaw	1	81	Wheel (only bigger axle).....	2
37	Hinge pin.....	1	82	Mudguard (only bigger axle).....	2
38	Chain lubrication plate.....	1	83	Middle plate.....	1
39	Bearing.....	2	84	Hydraulic pump (only tractor-driven).. 1	
40	Eye screw.....	1			
41	Spring.....	1			
42	Clamping bar for spring	1			

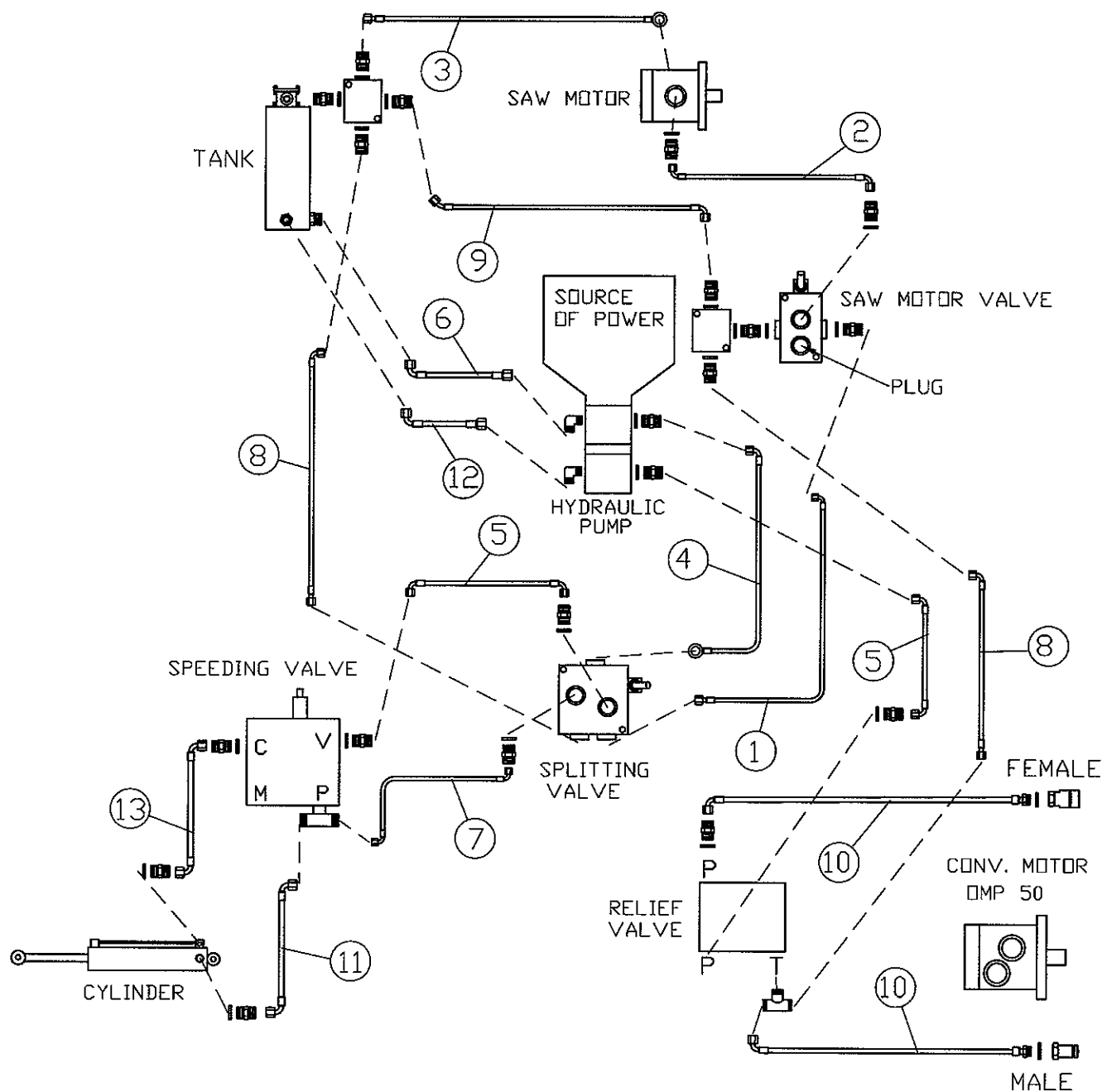
Japa 300 conveyor explosion view



Japa 300 conveyor spare parts list

Part	Part name	Pcs
1A	Lower frame	1
1B	Upper frame	1
2	Roller axle	1
3	Bearing.....	2
4	Upper end roller	1
5	Spring.....	2
6A	Upper side plate, left	1
6B	Upper side plate, right	1
7	Top end guard	1
8	Support.....	1
9	Chain for support.....	1
10	Belt.....	1
11	Belt scraper	7
12	Belt joining pieces	2
13	Bearing.....	2
14	Lower end roller	1
15	Clutch for shaft	1
16	Shaft for motor	1
17	Worm gear	1
18	Hydraulic motor	1
19	Flange for hydraulic motor.....	1
20	Belt holder	1
21	Lower end rubber	1
22	Joining for lower end rubber.....	1
23	Lower end protector	1
24	Flange for electric motor.....	1

Hydraulics diagram



Electric motor driven machine is equipped with only one hydraulic pump.

Osa	Osan tai kokoonpanoryhmän nimitys			Piiirustuksen ja osan n:o	Laatu	Muoto, mitat, malli				Kpl
					Aines					
					Osan lajimerkki					
Valmiste	Liittyy	Massa	kg	Toleranssittomat mitat		Suhde	Piirt.	RU	02.11.05	
		Laskettu					Tark.			
		Punnittu					Hyv.			
Laitilan Rautarakenne	JAPA 300 FIREWOOD MACHINE WITH HYDRAULIC CONVEYOR									
						Korvaa		Korvattu		